A failure of the DeTemple complete-monotonicity threshold

Research proof and Lean formalization

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Abstract

We refute the sufficient condition in Yang's complete-monotonicity conjecture, retained in the published paper of Yang, Chu and Zhang. The failure occurs for a positive parameter strictly inside the proposed range. The proof uses local continuity of the digamma function and a logarithmic midpoint obstruction. The literal source refutation, including the interior-parameter conclusion, is formalized in Lean.

For $x > 0$, let

$$\psi(x) = \frac{\Gamma'(x)}{\Gamma(x)}, \qquad R(x) = \psi(x + \tfrac{1}{2}) - \log x,$$

and use the exact function from [1, eq. (3.3)]

$$G_a(x) = 24\left(x^4 + ax^2 + \frac{7}{40}a - \frac{31}{336}\right)R(x) - \left(x^2 - \frac{7}{40} + a\right).$$

Conjecture 1(ii) of that paper states that $-G_a$ is completely monotone on $(0, \infty)$ if and only if $a \le 155/294$. Complete monotonicity means smoothness and $(-1)^n f^{(n)}(x) \ge 0$ for every $n \ge 0$ and $x > 0$; in particular it implies convexity.

Theorem 1. *There exist $0 < a < 155/294$ and $x > 0$ such that*

$$G_a(2x) < \frac{G_a(x) + G_a(3x)}{2}.$$

Consequently $-G_a$ is not convex on $(0, \infty)$ and is not completely monotone. Thus the conjectured sufficient condition is false.

The strict inequalities for both a and x are part of the conclusion. The ratio monotonicity question in Conjecture 1(i) is not settled here.

Lemma 1. *Let $c > 0$ and let h be real-valued and continuous at zero. Put*

$$F_c(x) = 24(x^4 + cx^2)(h(x) - \log x) - x^2, \quad x > 0.$$

For all sufficiently small positive x, $F_c(3x) - 2F_c(2x) + F_c(x) > 0$.

Proof. For $x > 0$, expanding and using $\log(kx) = \log k + \log x$ gives

$$\frac{F_c(3x) - 2F_c(2x) + F_c(x)}{x^2} = 24\big(B_c(x) - (50x^2 + 2c)\log x\big) - 2, \tag{1}$$

where

$$\begin{aligned} B_c(x) = {} & (81x^2 + 9c)h(3x) - 2(16x^2 + 4c)h(2x) + (x^2 + c)h(x) \\ & - (81x^2 + 9c)\log 3 + 2(16x^2 + 4c)\log 2. \end{aligned}$$

Continuity gives $B_c(x) \to 2ch(0) - 9c\log 3 + 8c\log 2$ as $x \downarrow 0$. Since $50x^2 + 2c \to 2c > 0$ and $-\log x \to +\infty$, the right side of (1) tends to $+\infty$. Multiplying by $x^2 > 0$ proves the claim. □

Proof of Theorem 1. Set $a_0 = 155/294$. The constant term in the polynomial multiplying R cancels:

$$\frac{7}{40}a_0 - \frac{31}{336} = 0.$$

The function $h(x) = \psi(x+1/2)$ is continuous at zero because Γ is analytic and nonzero near $1/2$. Therefore

$$G_{a_0}(x) = F_{a_0}(x) + \frac{7}{40} - a_0$$

satisfies the strict midpoint inequality for some $x > 0$ by Lemma 1. For this fixed x, the midpoint defect

$$a \longmapsto \frac{G_a(x) + G_a(3x)}{2} - G_a(2x)$$

is continuous (indeed affine) in a and is positive at a_0. It remains positive at some a with $0 < a < a_0$. Negating the inequality contradicts convexity of $-G_a$, proving both conclusions. □

Formal scope

The formal definition of ψ is the actual real derivative `deriv Real.Gamma / Real.Gamma`. Its local continuity is proved from complex Gamma analyticity, with the real derivative identification checked. The proof includes (1), its one-sided infinite limit, the strictly positive midpoint witnesses, continuity in the source parameter and the implication from complete monotonicity to convexity. The endpoint `DeTemple.source_counterexample_328867ii` is a premise-free negation of the literal source claim; `source_interior_not_completelyMonotone` also records $0 < a < 155/294$.

The certificate uses Lean 4.34.1 and mathlib at the pinned revision

`d13f23b723b8a846827a245b89c10fc7d3f11612`.

The separate rational point in the working mathematical notes is not asserted as a formal endpoint. This manuscript claims no optimal replacement threshold. Standard Gamma, convexity, continuity and logarithm facts are credited ingredients. No worldwide-priority claim is made.

References

[1] Z.-H. Yang, Y.-M. Chu and X.-H. Zhang, *Necessary and sufficient conditions for functions involving the psi function to be completely monotonic*, Journal of Inequalities and Applications **2015**, 157; doi:10.1186/s13660-015-0674-8. The conjecture also appears in Z.-H. Yang, arXiv:1409.6411 (2014), *Several completely monotone functions related to DeTemple's sequence*.